

NSE Stock Analysis: Insights Report

Six NSE stocks, 2015-01-01 to 2018-07-31 (889 trading days each). Moving averages, golden-cross signals, and the data problem hiding in the prices.

1. Summary

- **TVS Motors and Eicher Motors led the group.** Closing prices rose 86.9% and 82.6% from first to last trading day. The other four stocks gained far less (Section 4).
- **Two price events needed adjustment.** TCS on 2018-05-31 and Infosys on 2015-06-15 each show a close that roughly halved overnight and stayed at the new level, consistent with a 1:1 bonus issue. Dividing pre-event prices by 2 turns TCS from -23.8% to +52.4% and Infosys from -30.9% to +38.2% (Section 3).
- **The golden-cross rule produced 54 round trips across the six stocks; 21 made money.** Eicher Motors was the only stock where every round trip was a winner (6 of 6) (Section 4).
- **Quick reversals were common and costly.** 34 consecutive signal pairs were 30 calendar days apart or fewer. Of the 11 round trips that short, 10 lost money (Section 5).
- **Latest signals and long-run trend disagree for Eicher and TVS.** Both ended on a Sell despite the two largest gains. The crossover is a short-term momentum reading, not a verdict on the stock (Section 4.3).
- **The rule lags by construction.** A 50-day average needs 50 days of history, so no signal can appear in the first 10 weeks, and every signal confirms a move that has already begun (Section 6).

2. Data and method

Data. Daily closing prices for Bajaj Auto, Eicher Motors, Hero Motocorp, Infosys, TCS and TVS Motors, loaded from six CSV files into a database (dates converted to YYYY-MM-DD, empty cells to NULL). Only `close_price` is used. Each table has 889 rows. Each table has exactly one missing value in `deliverable_qty` and `pct_deliv_qty` (2017-08-31 for Bajaj and Infosys, 2015-12-09 for the other four); the same date missing across companies points to the exchange's reporting, and it does not affect this analysis.

Code. The submitted analysis is one file, `stock_market_analysis.txt`, written for MySQL 8. It drops and recreates the six tables, loads the CSV files, and then runs tasks 1-13 plus three extension tasks (14: TCS adjusted signals, 15: all-stock adjusted signals, 16: whipsaws and round trips). The file names such as `sql/12_worst_day.sql` cited below are the same queries, one file per task, used during development in SQLite. The MySQL file matched the SQLite results I compared (Buy, Sell and Hold counts, last signals, round-trip table).

Adjustment. For TCS and Infosys, close prices before the event date are divided by 2 (Section 3). The other four stocks use raw prices.

Signal rule. For each stock, MA20 and MA50 are the simple averages of the last 20 and 50 closes, left empty until a full window exists. **Buy** on the day MA20 crosses above MA50 (today $MA20 > MA50$ and yesterday $MA20 \leq MA50$); **Sell** on the day it crosses below; every other day is Hold. Averages are unrounded; rounding to 2 decimals gave the same Buy and Sell counts and the same last signals for all six stocks.

Round trips. A round trip is a Buy followed by the next Sell. Its return is $(\text{sell close} - \text{buy close}) / \text{buy close}$. Brokerage costs and dividends are excluded.

3. Two price events

3.1 TCS, 2018-05-31

Claim. TCS's close fell from 3517.75 to 1744.80 between 2018-05-30 and 2018-05-31 and did not return to the earlier level. The pattern is consistent with a 1:1 bonus issue (adjustment factor 2).

Evidence. `sql/12_worst_day.sql` ranks 2018-05-31 as TCS's worst day (-50.4%). The table below shows three trading days on each side.

Caveat. The event type is inferred from the price pattern, not from an external source. Adjusted figures depend on the factor-2 assumption.

Date	Raw close (Rs.)	Adjusted close (Rs.)
2018-05-28	3503.80	1751.90
2018-05-29	3522.70	1761.35
2018-05-30	3517.75	1758.88
2018-05-31 (event)	1744.80	1744.80
2018-06-01	1732.25	1732.25
2018-06-04	1744.55	1744.55
2018-06-05	1721.20	1721.20

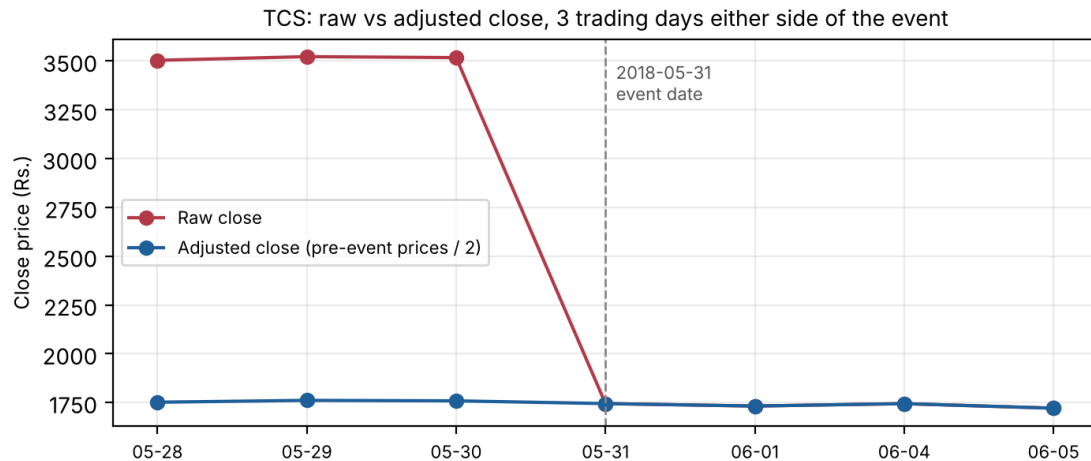


Figure 1. Raw series (red) has a cliff at the event date; the adjusted series (blue) is continuous. Plotted from the seven closes in the table above.

3.2 Infosys, 2015-06-15

Claim. Infosys shows the same pattern on 2015-06-15: the close fell from 1976.65 to 991.10 (-49.9%) and stayed near the new level, also consistent with a 1:1 bonus issue.

Evidence. `sql/12_worst_day.sql` (worst day); the table below shows three trading days on each side.

Caveat. Same as Section 3.1: the factor of 2 is assumed, not confirmed externally.

Date	Raw close (Rs.)	Adjusted close (Rs.)
2015-06-10	2026.20	1013.10
2015-06-11	2001.85	1000.92
2015-06-12	1976.65	988.33
2015-06-15 (event)	991.10	991.10
2015-06-16	999.45	999.45
2015-06-17	995.80	995.80
2015-06-18	1000.90	1000.90

3.3 What the adjustment changes

Percentage change, first to last close (`sql/11_pct_change.sql`, `sql/13_adjusted.sql`):

Stock	Raw	Adjusted
TCS	-23.8%	+52.4%
Infosys	-30.9%	+38.2%

Signals (`sql/15_adjusted_signals_all.sql`):

Stock	Series	Buys	Sells	Last signal
TCS	Raw	12	13	Sell, 2018-06-05
TCS	Adjusted	12	12	Buy, 2018-04-20
Infosys	Raw	9	9	Buy, 2018-05-07
Infosys	Adjusted	10	10	Buy, 2018-05-07

Dates where the raw and adjusted signals differ:

Stock	Date	Raw signal	Adjusted signal
TCS	2018-06-05	Sell	Hold
Infosys	2015-07-01	Hold	Buy
Infosys	2015-07-08	Hold	Sell
Infosys	2015-07-27	Hold	Buy
Infosys	2015-08-18	Buy	Hold

For TCS the raw series produces one extra Sell, three trading days after the event date, which the adjusted series does not. For Infosys, three of the four differences are signals that appear only in the adjusted series (the cliff hid them) and one is a raw Buy that disappears after adjustment. All five dates fall while the cliff is still inside the moving-average windows, which is the likely cause; a Buy or Sell that appears or disappears only because of the cliff is not information about the stock.

4. Per-stock results

4.1 Summary table

TCS and Infosys use the adjusted series. % change is first to last trading day (sql/11, sql/13); signals from sql/10 and sql/15; round trips from sql/16.

Stock	First close	Last close	% change	Buys / Sells	Last signal	Round trips	Winners	Avg return
TVS Motors	276.85	517.45	+86.9%	8 / 8	Sell, 2018-05-17	8	3	+11.2%
Eicher Motors	15239.15	27820.95	+82.6%	6 / 7	Sell, 2018-06-06	6	6	+10.2%
TCS (adj.)	1274.10	1941.25	+52.4%	12 / 12	Buy, 2018-04-20	11	2	-3.4%
Infosys (adj.)	987.90	1365.00	+38.2%	10 / 10	Buy, 2018-05-07	9	3	-0.4%
Bajaj Auto	2454.10	2700.70	+10.0%	12 / 11	Buy, 2018-06-21	11	4	+0.4%
Hero Motocorp	3107.30	3293.80	+6.0%	9 / 9	Sell, 2018-05-22	9	3	-1.7%

Avg return = arithmetic mean of each Buy-to-Sell return for that stock; it is not compounded and is not comparable with the first-to-last % change.

4.2 Stock by stock

Claim. TVS Motors rose 86.9%, the largest gain in the group.

Evidence. One round trip (2017-01-06 to 2018-01-29) returned +86.7% and dominates the +11.2% average; the other seven trips average +0.4% (sql/16).

Caveat. Eight round trips is a small sample and one trade drives the result. The rule uses prices only.

Claim. Eicher Motors rose 82.6%, and all 6 round trips were profitable.

Evidence. sql/11 and sql/16: 6 round trips, 6 winners, average +10.2%. Its highest closes (all above Rs. 32,000) were in September 2017 (sql/02).

Caveat. Six trades is too few to call this a reliable edge; the rule cannot detect a peak in advance.

Claim. TCS rose 52.4% on the adjusted series. The raw -23.8% is an artefact of the bonus event.

Evidence. sql/13 (adjusted change); sql/15 (raw last signal is a Sell on 2018-06-05, adjusted last signal is a Buy on 2018-04-20). Eleven round trips: 2 winners, average -3.4%.

Caveat. The adjusted last signal is 102 days before the end of the data, so it is old. The factor-2 assumption is not confirmed externally.

Claim. Infosys rose 38.2% on the adjusted series. The raw -30.9% is an artefact of the bonus event.

Evidence. sql/13; sql/15 (adjusted signals 10 / 10 against raw 9 / 9, differing on four dates in 2015). Nine round trips: 3 winners, average -0.4%.

Caveat. Factor-2 assumption as above. Signals in mid-2015 depend on the adjustment.

Claim. Bajaj Auto rose 10.0%, but its 11 round trips averaged only +0.4% with 4 winners.

Evidence. sql/10, sql/16. Five of the 11 round trips lasted 30 days or fewer and averaged -4.1%.

Caveat. Average return is not compounded, and several losses cluster in December 2015 and February 2018 (Section 5).

Claim. Hero Motocorp rose 6.0%; the rule produced 9 round trips with 3 winners and an average of -1.7%.

Evidence. sql/10, sql/16.

Caveat. Small sample; average return is not compounded.

4.3 Does the latest signal agree with the long-run trend?

Stock	Latest signal	Days before end of data	Full-period change	Agree?
Bajaj Auto	Buy, 2018-06-21	40	+10.0%	Yes
Eicher Motors	Sell, 2018-06-06	55	+82.6%	No
Hero Motocorp	Sell, 2018-05-22	70	+6.0%	Weakly (small gain)
TVS Motors	Sell, 2018-05-17	75	+86.9%	No
Infosys (adj.)	Buy, 2018-05-07	85	+38.2%	Yes
TCS (adj.)	Buy, 2018-04-20	102	+52.4%	Yes

Claim. For Eicher and TVS the signal and the long-run trend point in opposite directions. They answer different questions, so neither should override the other: the 3.5-year change says what the price did; a Sell means only that the 20-day average has fallen below the 50-day average, i.e. recent weeks are weaker than the preceding two months.

Evidence. Section 4.1 and the table above. Eicher's signals were profitable 6 of 6 times, TVS's only 3 of 8.

Caveat. Past signal quality does not predict the next one, and a Sell 55 to 75 days before the end of the data may already have reversed. This is not investment advice.

5. Whipsaws

Across all six stocks, 34 consecutive signal pairs were 30 calendar days apart or fewer (sql/16). 11 round trips (Buy then Sell) were that short, and 10 of the 11 lost money.

Stock	Signal pairs 30 days or fewer	Round trips 30 days or fewer	Average return of those trips
Bajaj Auto	11	5	-4.1%
TCS	8	1	-4.4%
Infosys	6	2	-3.3%
Hero Motocorp	4	1	-6.5%
TVS Motors	4	2	-8.3%
Eicher Motors	1	0	n/a

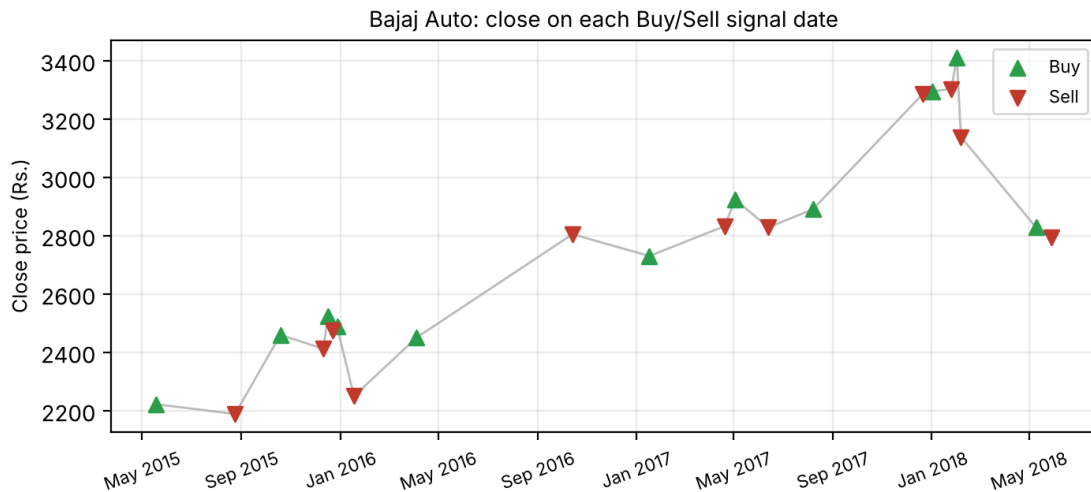


Figure 2. Bajaj Auto close on each of its 22 closed-out signals (11 Buys, 11 Sells), from the sql/16 ledger. The final open Buy on 2018-06-21 is not plotted.

5.1 Examples

Claim. Bajaj Auto, December 2015: four signals in 17 days.

Evidence. Sell 2015-12-11 (2411.90), Buy 2015-12-17 (2524.20), Sell 2015-12-23 (2475.60), Buy 2015-12-28 (2487.45). The Buy of 12-17 closed out at -1.9% six days later; the Buy of 12-28 closed out at -9.6% on 2016-01-18.

Caveat. One cluster in one stock; it shows what a sideways market does to a crossover rule, not how often it will happen.

Claim. Bajaj Auto, February 2018: a Buy on 2018-02-01 at 3409.50 was followed by a Sell on 2018-02-06 at 3138.20, a loss of 8.0% in 5 days.

Evidence. sql/16 round-trip table.

Caveat. Single trade.

Claim. TCS, October 2015: Buy on 2015-10-13 (1298.70, adjusted), Sell the next day (1241.70), -4.4%. This is the shortest gap in the data.

Evidence. sql/16 shortest-gap table.

Caveat. Single trade; the cause of the move is not in this data.

Five shortest-gap signal pairs:

Stock	First signal	Close	Second signal	Close	Gap (days)
TCS	Buy, 2015-10-13	1298.70	Sell, 2015-10-14	1241.70	1
Infosys	Sell, 2015-09-28	1108.20	Buy, 2015-10-01	1173.25	3
TCS	Sell, 2017-02-09	1162.13	Buy, 2017-02-13	1205.15	4
Bajaj Auto	Sell, 2015-12-23	2475.60	Buy, 2015-12-28	2487.45	5
Bajaj Auto	Buy, 2018-02-01	3409.50	Sell, 2018-02-06	3138.20	5

TCS closes are on the adjusted series.

6. Method questions

6.1 How early can a golden cross tell you anything?

A moving average uses only past closes. MA50 first has a value on trading day 50 (2015-03-13 here), and a crossover also needs the previous day's MA50, so no signal is possible before day 51, about ten weeks into the data. In practice Bajaj's first Buy came on 2015-05-18, 137 calendar days after the data starts. Because both averages trail the price, every signal confirms a move that has already happened; a sharp reversal takes several days to show up as a crossover.

6.2 Which conclusions in the course deck change after the adjustment?

Deck conclusion	After checking against the database and adjusting
TCS fell 23.8%	TCS rose 52.4%. The deck's figures also do not match the data (see below).
Infosys fell 3%	Infosys fell 30.9% unadjusted and rose 38.2% adjusted.
TCS: 12 buys, 13 sells	12 buys, 12 sells on the adjusted series (the 2018-06-05 Sell was caused by the cliff).
Infosys: 9 buys, 9 sells	10 buys, 10 sells on the adjusted series.
TCS trend shifting down; TCS on the sell list	TCS's last adjusted signal is a Buy (2018-04-20).
Eicher +82.5%	+82.6% (rounding).

Deck figures that do not match the data: the deck lists TCS as 2548.20 on 2018-07-31 and 2454.1 on 2015-01-01. In the data, 2548.20 is the first close (2015-01-01) and the last close is 1941.25; 2454.1 is Bajaj Auto's first close. The deck's other price-change figures for Bajaj, TVS and Hero match the database.

6.3 What this data cannot tell you

Missing	Why it matters
Dividends	Total return is higher than the price return, so this analysis understates long-run gains.
Brokerage and taxes	Every round trip has costs; thin returns on short trades may not survive them.
A market benchmark	Without an index such as the Nifty 50, there is no way to say whether the rule beats simply holding.
News and fundamentals	Prices alone cannot separate a correction from a structural decline, or explain a market-wide drop such as 2015-08-24, the worst day for both Bajaj and Eicher.

7. Limits of this analysis

- **Small samples.** Each stock has 6 to 11 round trips. One unusual trade can dominate an average.
- **Averages are not buy-and-hold.** The average round-trip return is a simple mean of per-trade returns, not compounded, and is not comparable with the first-to-last % change.
- **TVS Motors outlier.** One round trip returned +86.7%; the other seven average +0.4%.
- **No out-of-sample test.** The signals were computed and described on the same data; nothing here shows they would work in another period.
- **Event adjustment.** The TCS and Infosys adjustments assume a clean 1:1 bonus issue (factor 2), inferred from the price pattern.

Source: the six NSE CSV files; queries in stock_market_analysis.txt (tasks 1-16, MySQL 8; sql/ holds the per-task SQLite versions). Core figures were cross-checked with src/verify_report.py.